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Which one is the true statement regarding systems with web-services?

Select one:

- a. Supports dynamic integration of components
- b. Use proprietary protocols for communication HTTP, Json
- c. Web services always use monolithic architecture
- d. Web services are about user to program interaction program to prrogram

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Which language supports both procedural and object oriented programming?

Select one:

- a. Java Object-Oriented language
- b. C procedural language
- c. C++
- d. C# Object-Oriented language

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The CRUD operations are determined by?

1. Read & Create
2. PUT & DELETE
3. HTTP verb
4. All of them

HTTP verbs (or methods, as they are properly called) are POST, GET, PUT, PATCH, and DELETE.

These correspond to create, read, update, and delete (or CRUD) operations

PUT - creates a new resource or replaces a representation of the target resource with the request payload.

Select one:

- a. Only (3)
- b. Only (4)
- c. Only (2)
- d. Only (1) and (2)

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Which of the following is not a software category?

Select one:

- a. CASE tools — Computer-Aided Software Engineering Tools
- b. Utilities
- c. System software
- d. Cloud computing — Cloud computing is the service, not a software

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Identify correct ways to maintain the quality of the code:

1. Documenting the code
2. Functional testing
3. Regular code reviews
4. Clear requirements

Select one:

- a. Only (2) and (3)
- b. Only (1) and (3)
- c. Only (2) and (4)
- d. All (1), (2), (3), and (4)

Evaluate WSDL

Select one:

- a. All of others
- b. WSDL helps in dynamic service integration
- c. WSDL is a contract between the Web service and the client
- d. WSDL describes a web service

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What is the verb, which we use to indicate Create operation? (1) GET (2) POST (3) PUT (4) DELETE

Select one:

- a. Only (2)
- b. Both of them
- c. None of them
- d. Only (3)

GET =>> Read
PUT =>>Update/Replace
PATCH =>> Update or Modify
DELETE =>> Delete

HTTP REQUEST Methods-GET, POST, PUT, PATCH, DELETE

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What is not related to the scalability?

Select one:

- a. Distribution of the components
- b. Testing
- c. Number of users
- d. Concurrent access

This refers to many users or processes hitting the system at the exact same time.

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Some applications developed using particular programming languages can be directly run on the OS witho

Select one:

- a. This statement is true
- b. Cannot validate the statement without knowing the OS
- c. This statement is not conclusive
- d. This statement is false

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What is false about Micro-services Architecture?

Select one:

- a. The system is developed with many components
- b. Can use single database or multiple databases
- c. May use SOAP or REST for communication
- d. Made of many web services

components:

Clients
Identity Providers
API Gateway
Messaging Formats
Databases
Static Content Management
Service Discovery

microservices approach offers tangible benefits including an increase in scalability, flexibility, agility, and other significant advantages.

Keep a separate database for each service

REST for microservices is that services can communicate without requiring internal knowledge of one another.

You can use a single shared database with tables that owned by different microservices if your data is heavily related. Also in case you have strong requirements on data consistency and availiblty of a service.

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What are the languages, which can be used to write code in an HTML page, which can

Select one:

- a. CSS, Java
- b. None of the above
- c. JS, Java
- d. CSS, JS

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The communication techniques in distributed systems can be categorized as

Select one:

- a. RPC, RMI, and CORBA
- b. Functional oriented, Message oriented, and Resource oriented
- c. RPC/RMI, SOAP, and DC
- d. Delta-Communication, SOAP communication, and RESTful communication

Functional oriented communication
- RPC/RMI
- CORBA

2. Message Oriented communication
- SOAP

3. Resource oriented communication
- REST

4. Communication in Rich Web-based Applications

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Refactoring is one of the aspect of quality code. What does it mean?

Select one:

- a. Re-arrange the code to improve efficiency
- b. Manage the configurations
- c. Extending the code
- d. Test and fix the bugs

refactoring is the process of restructuring existing computer code without changing its external behavior.

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Web-based applications can be categorized into

Select one:

- a. standalone applications and client-server applications
- b. standalone applications and browser-based applications
- c. browser-based applications and non-browser-based applications
- d. web applications and browser-based applications

Browser-based applications: Chrome, Firefox, or Safari
Non-browser-based applications: Instagram or Spotify

Standalone applications: These are typically software that works offline without needing a network connection

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What are the benefits of web services?

1. Static integration
2. Standardized Protocol HTTP, XML, and SOAP) to communicate.
3. Interoperability It allows different applications written in different languages to talk to each other seamlessly.
4. Constant by design
5. Low Cost of communication

Select one:

- a. 1 and 5
- b. 3, 4, and 5
- c. 2, 3, and 5
- d. 1, 2, and 4

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Procedural programming focuses on

Select one:

- a. Flow control with GOTO statements
- b. Use of namespaces/packages
- c. Functions (as in functional programming)
- d. Modularity and code reuse

Modularity: It allows you to break down a large program into smaller, manageable, and logical "modules" or procedures.

Once a procedure is written, it can be called multiple times throughout the program,

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Which cannot be used to develop the server-model?

Select one:

- a. PHP
- b. HTML
- c. Node.JS
- d. JAVA

Most large-scale websites use server-side code to dynamically display different data when needed, generally pulled out of a database stored on a server and sent to the client to be displayed via some code (e.g. HTML and JavaScript).

•To develop the server model, a server-side language/framework and related technologies should be used.

•PHP, JAVA, .NET, Python,

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The difference between the Servlets and the JSP:

Select one:

- a. Servlets are used to develop web applications and JSP is used to develop web sites
- b. JSP is an application and Servlet is a web page
- c. Servlet can generate HTML and JSP can generate JAVA code
- d. JSP is a web page and Servlet is an application

JSP – Web pages with Java code
•Servlets – Java applications, which generate HTML

Server-side - frameworks and libraries
• JAVA – JSP/Servlet

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Which one of the following is not based on GIT?

Select one:

- a. SourceForge
- b. GitHub
- c. OneDrive
- d. GitLab

gitBucket
GitHub
GitLab
SourceForge
cloudForge

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The resource is determined by? (1) The domain name (2) CRUD operations (3) URL segments

Select one:

- a. None of them
- b. Only (3)
- c. Only (2)
- d. Only (1)

CRUD = HTTP verbs

Domain name :

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Which one of the following is not a benefit of CDN?

Content Delivery Networks

Select one:

- a. Reducing bandwidth costs
- b. Improving the security of the system
- c. Support collaboration
- d. Improving system's loading time

Increasing content available and redundancy

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What modules are implemented on the server-side of BAW-MVC?

1. Component
2. Controller
3. View
4. Model

Select one:

- a. Only (1) and (2)
- b. Only (1), (2), and (3)
- c. Only (3), and (4)
- d. All of them

The Server (Model/View) is the Kitchen. It holds the ingredients and prepares the plate.

The Browser (Controller) is the Waiter. The waiter stays "on the floor" (the client side) to hear your order immediately and then communicates back to the kitchen when necessary.

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What are the advantages of MVC?

Select one:

- a. Presentation is separated from the business logic
- b. All of the others
- c. Event-handling is separated from business logic
- d. Presentation is separated from Event-handling

Simultaneous Development: Multiple developers can work on the View and the Model at the same time without interfering with each other.

High Cohesion: Logic is grouped logically by its purpose.

Ease of Modification: You can change the entire look of an application (the View) without touching the underlying business logic (the Model).

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Programming languages provide

Select one:

- a. Semantics and functions
- b. Syntax and procedures
- c. Syntax and functions
- d. Syntax and semantics

Syntax : Rules (forgetting a semicolon in Java is a syntax error)

Semantics: This refers to the meaning and logic of the code.

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Bootstrap is a?

Select one:

- a. Scripting language
- b. Plugin
- c. Framework
- d. Library

Frontend Framework

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CSS can be written as

Select one:

- a. all of other answers
- b. Inline CSS
- c. External CSS sheets
- d. Internal CSS sheets

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Which one of the following is not a VCS model?

Select one:

- a. Centralized version control systems
- b. Local version control systems
- c. Distributed Version Control Systems
- d. Global version control systems

Version Control Systems(VCS) is of 03 types : Centralized VCS
Local VCS
Distributed VCS

Why do centralized control in distributed systems useful to engineers?

Select one:

- a. Maintaining the system is easier
- b. All of the others
- c. Easier to update
- d. Easier to modify and extend

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What is NOT a correct statement about CSS?

Select one:

- a. None of the others
- b. Eliminate HTML formatting tags ✓
- c. Save work and time by reusing style sheets ✓
- d. Slow down loading

CSS is Fast Loading

As the stylesheets can be used again again instead of repeating the same formating tasgs.

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What is the operation for HTTP verb PUT? (1) Create (2) Update

Select one:

- a. None of them
- b. Only (1)
- c. Only (2)
- d. Both of them

POST =>> Create
GET =>> Read
PUT =>>Update/Replace
PATCH =>> Update or Modify
DELETE =>> Delete

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Which of the following is incorrect about JAX-RS annotations?

Select one:

- a. PATH - Relative path of the resource class/method
- b. POST - Create a new resource
- c. PUT - Used to add resources
- d. HEAD - Get status about method availability → used to retrieve the metadata

•JAX-RS is an API
•There are two implementations

jersey

RESTeasy

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Which of the following is not correctly defining loosely coupled architecture of web service?

Select one:

- a. services can be independently updated
- b. software systems become more manageable
- c. allows integration between different systems
- d. Low scalability

one of the primary reasons for choosing a loosely coupled architecture is to achieve high scalability.

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"In software runtime architecture we may need additional virtualization layers, depending on the type of the system." Between which layers we may need these virtualization layers?

Select one:

- a. Between Operating System and Hardware
- b. Between User and Application
- c. Between User and Hardware
- d. Between Application and Operating System

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the Model-View-Controller (MVC) architectural style, the model defines the

elect one:

- a. Interfaces T
- b. Presentation T
- c. Business-logic
- d. Communication model x

Model: This is the heart of the application. It manages the data, database interactions, and the business logic

View: This handles the presentation. It is what the user sees on the screen (the UI).

Controller: This acts as the interface

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The key technique in RESTful communication uses: (1) URL (2) HTTP Verbs

Select one:

- a. Only (1)
- b. None of them
- c. Only (2)
- d. Both of them

RESTful communication relies on:

URLs → to identify resources (e.g., /users, /orders/1)

HTTP Verbs → to define actions on those resources:

GET → retrieve data

POST → create

PUT/PATCH → update

DELETE → remove

*There is a gap between development and implementation environments, such as having different platfo

Which one of the following helps us to overcome these issues?

Select one:

- a. Version controlling
- b. CDN
- c. Dependency management
- d. Virtualization

Virtualization allows you to package the entire environment—the OS, dependencies, and configurations—into a single image.

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What are the example for data representational elements in HTML?

Select one:

- a. Tables, Input
- b. Lists, Tables
- c. Input, Buttons
- d. Buttons, Tables

•Data representational elements
(these elements use nested structures)
• Lists
• Tables

•>>Form elements
• Input
• Radio buttons, check boxes
• Buttons

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What is used by Maven to store project details

Select one:

- a. CDN
- b. VCS
- c. XML file
- d. JSON file

Maven uses an XML file to store project details, specifically the pom.xml

Which is NOT correct about the View development in web-based applications?

Select one:

- a. View is deployed to the client's computer
- b. View should be loaded to the client's browser
- c. Server-side code in View may execute in server-side
- d. View should communicate with model to load the data from database

It should never communicate directly with the database or even with the Model to fetch data.

View should only communicate with the Controller.

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Which one of the following is not a purpose of having a VCS?

Select one:

- a. Restoring previous versions — If you make a mistake, you can "roll back" to a known working state.
- b. Backup — By pushing code to a remote repository
- c. Code formatting
- d. Collaboration — A VCS allows multiple developers to work on the same project simultaneously without overwriting each other's work through branching and merging.

Collaboration
- With a VCS, everybody on the team is able to work absolutely freely - on any file at any time
- The VCS will later allow you to merge all the changes into a common version

- Storing versions properly
- A version control system acknowledges that there is only one project

- Restoring previous versions

- Understanding what happened
- Every time you save a new version of your project, your VCS requires you to provide a short description of what was changed

- Backup

n 13

t answered

d out of

g question

The basic architecture of the distributed systems is called

Select one:

- a. Client-server architecture
- b. N-tier architecture
- c. 3-tier architecture
- d. Service oriented architecture

The basic architecture of the distributed systems is called the client-server (or two-tier) architectur

one or more clients (requestors) connect to a central server (provider) to access shared resources or services.

What HTTP verb is used for Read operation?

(1) GET

(2) POST

(3) PUT

(4) DELETE

POST =>> Create
GET =>> Read
PUT =>>Update/Replace
PATCH =>> Update or Modify
DELETE =>> Delete

Select one:

- a. (1) and (2) only
- b. All of them
- c. Only (1)
- d. only (2)

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A framework is incomplete?

Select one:

- a. The correct answer is not given
- b. Not conclusive
- c. Agree
- d. Disagree

A Framework is incomplete

• Framework itself is not usable on its own. Also they do not leave some fill-in-the-blanks for its user.

• Framework alone will not work, the relevant application logic should be implemented and be deployed along with the framework

• Frameworks pose trade-off challenge between learning curve and saving-coding-time

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A framework is incomplete?

Select one:

- a. The correct answer is not given
- b. Not conclusive
- c. Agree
- d. Disagree